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# Modeling system reliability aspects of ERTMS/ETCS by fault trees and Bayesian networks

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ABSTRACT: Critical control systems require proper techniques to predict their failure rate since early design stages, in order to fulfil dependability requirements and minimize development costs. Bayesian Networks have been shown to be suitable to model system reliability aspects, extending the modeling power of Fault Trees and featuring a better solving efficiency with respect to Petri Nets. In this paper we exploit the Fault Tree and Bayesian Network formalisms in order to perform a hardware reliability analysis of a complex real world case study: the European Railway Traffic Management System/European Train Control System (ERTMS/ETCS). ERTMS/ETCS is a recent standard specification aimed at improving interoperability, performances and dependability of modern railways. An implementation of ERTMS/ETCS is a distributed heterogeneous system with strict reliability requirements. Starting from such requirements and from a reference hardware architecture, we studied system reliability by instantiating models with realistic parameters and performing a series of sensitivity analyses in order to highlight design trade-offs. By evaluating and integrating sub-models using a compositional approach we both obtained several interesting results and showed the effectiveness of a combined use of Fault Trees and Bayesian Networks in dealing with system reliability analyses of train control systems.

## 1 INTRODUCTION AND RELATED WORKS

International RAMS (Reliability Availability Maintainability Safety) standards, e.g. CENELEC (CENELEC 1999), specifically address the techniques used to evaluate system dependability for critical control systems, giving general guidelines. The use of formal modeling techniques is highly recommended by such standards. Formal methods allow to predict system reliability since early stages of system development, reducing the probability of design reviews. The modeling language should be chosen with the aim of balancing easy of use, expressive power and solving efficiency. Several formal languages and methods have been proposed by the scientific community in order to model system reliability aspects. Among them, Fault Trees (FT) and Reliability Block Diagrams (RBD) are limited in expressive power, but they are very efficient and easy to use; Continuous Time Markov Chains (CTMC) and the various kinds of Stochastic Petri Nets (SPN) allow to model any complex structure or behavior, but are usually not compatible with the complexity of very large systems, as their solving algorithms suffer from the state space explosion problem. This is also true for Fault Tree extensions which are solved by translating the extended FT model into a CTMC or SPN model (e.g. Dynamic and/or Repairable Fault Trees, see (Dugan 1992) and (Flammini 2005)). Finally, Bayesian Networks (BN) have been recently shown to be able to balance expressive power and solving efficiency in order to model system reliability aspects. BN and their extensions (e.g. Dynamic Bayesian Networks, see (Montani 2005)) provide a unified framework which is able to model nearly all reliability related issues. BNs have also been show to augment the expressive power of Fault Trees and achieve better efficiency with respect to Petri Nets and their extensions (e.g. GSPN, see (Ajmone 1995)). The BN formalism supports multi-state events, noisy gates, common mode failures, decision extensions and can be used to detect reliability bottlenecks and to diagnose failure causes starting from observable symptoms (the "evidence"). A methodology to translate a FT into a BN and a performance comparison among FT, PN and GSPN are presented in (Bobbio 2001b) and (Bobbio 2001a) respectively.

ERTMS/ETCS (European Railway Traffic Management System/European Train Control System) is a European standard specification aimed at improving safety, reliability, performance and interoperability of European railway lines (UNISIG 2002). In this paper, we exploit the Fault Tree and Bayesian Networks

formalisms to model system reliability aspects of ERTMS/ETCS. To the best of our knowledge, no system level reliability study about ERTMS/ETCS has been performed in the research literature and no complex real-world case-study has been modeled in terms of its system level reliability using Bayesian Networks. In this work, the SHARPE (Sahner 1996) software package has been employed to solve FT models,, and Netica by Norsys (Netica 2006) has been used to draw and solve BN models. The case-study described in this paper is complex enough to show the effectiveness of the modeling approach, and to perform a reliability study, useful to better understand the coherence and reachability of the ERTMS/ETCS RAM requirements and how to size reliability parameters in order to meet them. We only marginally address the problem of criticality indices (NASA 2002), as it is not in the main scope of this work to locate and remove system reliability bottlenecks.

The paper is organized as follows: Section 2 presents a brief overview of the ERTMS/ETCS case-study; Section 3, 4, and 5 contain the Fault Tree Analyses and a discussion of results related to the Lineside, On-board and Trackside subsystems of ERTMS/ETCS respectively; Section 6 provides the description and evaluation of the global Bayesian Network model of the system, exploiting the results obtained by solving FT submodels described in previous sections; finally, Section 7 gives a brief summary of results and hints about future work.

## 2 THE ERTMS/ETCS CASE-STUDY

### 2.1 *Description of the ERTMS/ETCS system*

ERTMS/ETCS provides the specification of an On-board, a Lineside and a Trackside system. In this paper we consider ERTMS/ETCS Level 2 (L2), based on a fixed-block and continuous radio-signalling system. System architecture and main data flows are depicted in Figure 1.

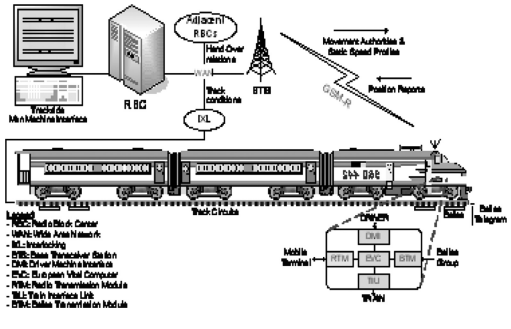



Figure 1. Architectural scheme and data flows of ERTMS/ETCS Level 2.

The Lineside system is distributed along the track. It consists of a set of Balise Groups (BG), each one made up by one or more balises. A balise is a device installed between rail-lines, which has the aim of transmitting data telegrams to the trains passing over it (data telegrams contain geographical positioning information). The On-board system is installed on the train. It is in charge of controlling train movement against a permitted speed profile (also known as braking or protection curve), which is elaborated using the information received from the Trackside via the GSM-R radio network. The On-board also communicates the position of the train, detected by reading balise telegrams, and other data (e.g. operating mode) to the Trackside, via Position Report radio messages. In order to perform train protection, the On-board must be equipped with the following devices:

- RTM (Radio Transmission Module): it provides a communication interface with the Trackside, using a GSM-R Mobile Terminal;
- BTM (Balise Transmission Module): it energizes balises and reads their telegrams;
- TIU (Train Interface Unit), used to interface with train borne apparels (e.g. emergency brakes);
- DMI (Driver Machine Interface): it provides on-board interaction with train driver for manual procedures;
- EVC (European Vital Computer): it provides the on-board control logic.

The EVC is an embedded, real-time and safety-critical computing system, so we will suppose that it is based on the well-known and highly adopted Triple Modular Redundant (TMR) architecture (2 out of 3 voting on processor outputs). In order to control train movement, the EVC has to interface with the on-board Odometer, measuring train speed and distance since last balise (which provides the train with its exact position, thus recalibrating the Odometer).

The Trackside mainly consists of Radio Block Centres (RBCs), which have the responsibility of providing trains with Movement Authorities (i.e. the distance they are allowed to move on), Static Speed Profiles (i.e. speed limitations) and possible emergency information. In order to detect the status of the track, the RBC needs to collect data coming from the national Interlocking (IXL) system. IXL is not object of standardization and for this reason its analysis is out of the aims of this paper. The RBC needs a safety-critical elaboration subsystem (let us suppose a TMR system, like the one of the EVC), and two main communication interfaces of the following types:

- GSM-R, in order to communicate with trains in its (limited) supervised area;
- WAN (Wide Area Network), used to interface with IXL, which is distributed along the track, and with adjacent RBCs.

### 2.2 *System RAMS requirements*

ERTMS/ETCS RAMS requirements define the (non functional) dependability related indices which a system implementation must satisfy in order to be fully compliant to the standard (UNISIG 1999). The study of the Safety part is not in the scope of this work, so we will consider only RAM specification. Furthermore, we do not consider performability indices (e.g. train delays, transmission errors, etc.), related to specific hardware performance and software implementation; we will only consider structural reliability aspects.

ERTMS/ETCS defines three main types of system failure modes:

- Immobilizing Failure (IF): a failure which causes two or more trains to be switched into on-sight mode (i.e. they are no more under full system supervision);
- Service Failures (SF): a failure which causes at most one train to be switched into on-sight mode;
- Minor Failure (MF): a failure that results in unscheduled maintenance and cannot be classified in the above defined failure conditions.

For each of these failure modes, RAM specification defines the required reliability indices (e.g. MTBF, Mean Time Between Failures) and the contribution coming from different parts of the system or abstraction levels (e.g. software vs hardware, Trackside vs On-board, etc.). Besides system level indices, also constituent level indices are indicated, so that designers can choose between a global approach to achieve system reliability and a more conservative approach, based on using more reliable and expensive components. Thus, the challenge consists in demonstrating compliance to system level RAM requirements using less reliable constituents. Table 1 summarizes the most important indices that will be considered in the following of this paper (a bracketed asterisk corresponds to a constituent level requirement).

Table 1. ERTMS/ETCS RAM requirements of interest.

| Param. | Description | Value |
|---|---|---|
| MTBF-$I_{ONB}$ | MTBF w.r.t. IFs due to the On-board | $>2.7*10^6$ h |
| MTBF-$S_{ONB}$ | MTBF w.r.t. SFs due to the On-board | $>3*10^5$ h |
| MTBF-$I_{TRK}$ | MTBF w.r.t. IFs due to the Trackside | $>3.5*10^8$ h |
| MTBF-$I_{LNS}$ | MTBF w.r.t. IFs due to the Lineside | $>1.2*10^5$ h |
| $U_{RBC}$ | RBC Unavailability (*) | $<10^{-6}$ |
| $U_{BAL}$ | Balise Unavailability (*) | $<10^{-7}$ |
| $A_{IF-HW}$ | System Availability w.r.t. hardware IF | $>0.999985$ |
| $A_{SF-HW}$ | System Availability w.r.t. hardware SF | $>0.99987$ |

## 3 THE LINESIDE SUBSYSTEM

### 3.1 *Lineside model structure*

The Lineside subsystem at ERTMS/ETCS L2 is not implementation specific in its hardware architecture, so the considerations presented in this section are very general. Once known track related parameters (i.e. track length and BG interdistance, which impact on total BG number), we can only act on the dimension of reliability related parameters (i.e. MTBF, MTTR and redundancy). In particular, the redundancy degree implies the number of balises for each group, which can vary from 2 to 8, according to the specification (we explicitly neglect the case of single balise groups, as they do not allow to detect train direction). Finally, we remark that ERTMS/ETCS RAM specification for constituents requires $U_{BAL} < 10^{-7}$. In our analysis, we provided a variation interval for Lineside reliability parameters in order to show that less reliable balises used in redundant groups are able to easily meet system level availability requirements at a less cost. In particular, we assumed the Lineside is responsible for an Immobilising Failure whenever two adjacent BGs fail, since this event causes the train to apply the emergency brakes as the so called "balise linking error reaction". The Fault Tree model for the Lineside is depicted in Figure 2 (the BG structure, being the same for all groups, is reported only once).

### 3.2 *Lineside model parameters*

The description and variability interval for parameters is reported in Table 2. The variability interval of the total number of balise groups has been chosen considering: (a) realistic track lengths from 100 Km to 400 Km, (b) an average BG inter-distance of about 1 Km, and (c) both track directions.

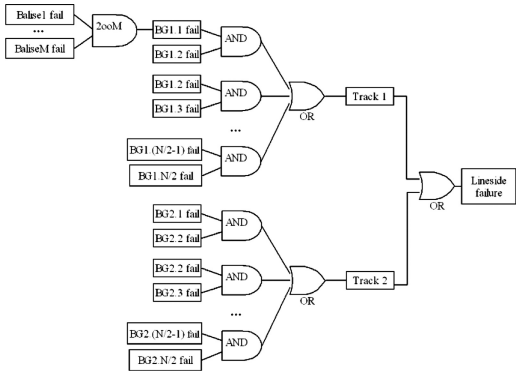



Figure 2. Fault Tree model of the Lineside subsystem.

Table 2. Lineside model parameters.

| Param. | Description | Min | Max | Step |
|---|---|---|---|---|
| $M_{BG}$ | Number of balises for each group | 2 | 8 | 1 |
| $N_{BG}$ | Total number of balise groups | 200 | 800 | 200 |
| $MTBF_{BAL}$ | Mean time between failures for balise [h] | $2.5\ 10^5$ | $1.5\ 10^6$ | $2.5\ 10^5$ |
| $MTTR_{BAL}$ | Mean time to repair for balise [h] | 0.5 | 2.5 | 1 |
| $U_{IF-LNS}$ | Lineside system unavailability with respect to IF | | | |

Table 3. A selection of Lineside results.

| $M_{BG}$ | $N_{BG}$ | $MTBF_{BAL}$ | $MTTR_{BAL}$ | $U_{IF-LIN}$ |
|---|---|---|---|---|
| 2 | 200 | $2.5\ 10^5$h | 0.5 h | $3.1840*10^{-9}$ |
| | | | 1.5 h | $2.8655*10^{-8}$ |
| | | | 2.5 h | $7.9598*10^{-8}$ |
| | | | 2.5 h | $1.9900*10^{-8}$ |
| | | $10^6$ h | 2.5 h | $4.9750*10^{-9}$ |
| | 400 | $7.5\ 10^5$ h | 1.5 h | $6.3839*10^{-9}$ |
| | 800 | $2.5\ 10^5$ h | 0.5 h | $1.2784*10^{-8}$ |
| | | | 2.5 h | $3.1959*10^{-7}$ |
| | | $5.0\ 10^5$ h | 0.5 h | $3.1960*10^{-9}$ |
| | | $10^6$ h | 1.5 h | $7.1909*10^{-9}$ |
| ≥3 | Any | Any | Any | $\approx 0\ (\ll 10^{-10})$ |

## 4 LINESIDE MODEL EVALUATION

Selected results of the analyses of the model in Figure 2 are shown in Table 3 (as aforementioned, the only significant failure mode for the Lineside leads to an Immobilising Failure). Table 3 suggests that BGs constituted by more than 2 balises are over-dimensioned with respect to ERTMS/ETCS availability requirements: such result formally justifies the practical choice of adopting groups constituted by just two balises in all current projects. The possibility to adopt BG of up to 8 balises seems therefore completely useless, as the only reason to do this would be using very low reliable balises, which is obviously not convenient, as frequent on-the-track interventions are difficult and costly. As for the other results, almost any combination of Lineside parameters produces acceptable results, with most of them leading to $U_{IF-LNS} < 10^{-8}$ (not all are shown in the table). The only results requiring attention are the ones corresponding to the worst combinations of parameters: maximum track length, lowest balise reliability, highest time to repair: even in such worst conditions, the result of $U_{IF-LNS} \approx 10^{-7}$ is perfectly compatible with the order of magnitude of the other ERTMS/ETCS subsystems, as it will be shown in the following sections. In fact, other ERTMS/ETCS subsystems (e.g. EVC, RBC, etc.) feature a similar unavailability, but in a typical installation they are usually required in a number which is more than one. However, the mentioned worst case corresponds to a balise unavailability:

$$U_{BAL}=1-MTBF_{BAL}/(MTBF_{BAL}+MTTR_{BAL})=$$
$$=1-2.5*10^5/(2.5*10^5+2.5)=10^{-5}$$

which is two orders of magnitude higher than the $10^{-7}$ value stated by RAM specification for constituents (see Table 1), thus justifying the convenience of a system level approach. Finally, the Lineside results presented above justify the possibility to neglect the Lineside subsystem contribution in a global system reliability analysis when a proper choice of parameters is performed.

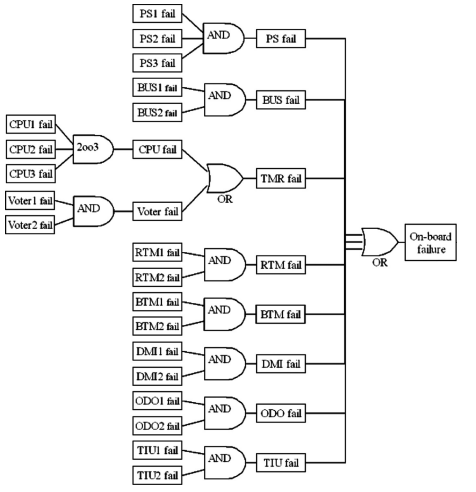



Figure 3. Fault Tree model of the On-board subsystem.

## 5 THE ON-BOARD SUBSYSTEM

### 5.1 *On-board model structure*

We will realistically assume the On-board system is not repairable on-line, for the unavailability of an on-board technician. Moreover, each On-board system only features a failure mode related to availability. In other words, at any time the On-board can only assume two states: available (working in full operating mode) and unavailable. A Fault Tree model based on components' MBTF perfectly fit the required analysis. The FT model comprises On-board components described in Section 2 in redundant configurations ("no single point of failures"), plus the ones constituting the

EVC elaboration subsystem (based on a basic TMR architecture). In particular, the EVC features: 3 CPU cards with dedicated memory; a redundant FPGA-based majority voter on CPU outputs; 3 redundant Power Supplies (PS); a system BUS interconnecting all the peripherals. All ERTMS/ETCS components are essential for correct on-board operation, and thus are connected to the Top Event of the Fault Tree via an OR gate. The FT model for the On-board is depicted in Figure 3: as it is quite self-explaining, we are not going to describe model details. In order to cause an Immobilising Failure, at least two on-boards must fail, while a single On-board failure implies a Service Failure.

### 5.2 *On-board model parameters*

For the EVC, Commercial Off The Shelf (COTS) components have been chosen. Parameter values are taken from typical component datasheets and should be considered only as orders of magnitude. Power Supply is chosen to be redundant twice, because it is usually less reliable than other components. For standard ERTMS/ETCS devices (e.g. BTM, RTM, etc.), the basic MTBF values have been chosen in accordance with specified RAM requirements for constituents. The chosen parameters are reported in Table 4. With safe train headways of at least 15 Km (considering train braking distance at a maximum speed of 300 Km/h), the average number of trains does not exceed 24 for typical track lengths (however, lower values are far more probable, as high-speed railway lines are not so heavily loaded).

Table 4. On-board model parameters.

| Parameter | Description | Value |
|---|---|---|
| $N_{ONB}$ | Total number of On-board systems | 2-24 |
| $MTTR_{ONB}$ | MTTR of the On-board | 30′, 1 h, 2 h |
| $MTBF_{CPU}$ | MTBF of the Processor-Memory Card | $1.35*10^5$ h |
| $MTBF_{BUS}$ | MTBF of system Bus | $2.25*10^5$ h |
| $MTBF_{VOT}$ | MTBF of each FPGA based Boter | $3.33*10^8$ h |
| $MTBF_{PS}$ | MTBF of Power Supply | $5.50*10^4$ h |
| $MTBF_{RTM}$ | MTBF of the Radio transmission module | $10^6$ h |
| $MTBF_{BTM}$ | MTBF of the Balise transmission module | $10^8$ h |
| $MTBF_{ODO}$ | MTBF of the On-board odometer | $10^7$ h |
| $MTBF_{TIU}$ | MTBF of the Train interface unit | $10^7$ h |
| $MTBF_{DMI}$ | MTBF of the Driver machine interface | $10^7$ h |
| $MTBF_{ONB}$ | MTBF of a single on-board system | |
| $MTBF_{IF-ONB}$ | MTBF of the On-board system with respect to IF | |
| $MTBF_{SF-ONB}$ | MTBF of the On-board system with respect to SF | |

Table 5. Results of the On-board sensitivity analysis.

| Scale | $MTBF_{ONB}$ |
|---|---|
| 10 | $6.4776*10^4$ |
| 1 | $6.4769*10^4$ |
| 0.1 | $6.4195*10^4$ |
| 0.01 | $4.4182*10^4$ |
| 0.001 | $0.6885*10^4$ |

### 5.3 *On-board model evaluation*

The results of On-board model evaluation have been obtained fixing COTS' MTBF values (which are given by their specification) and varying the MTBF of ERTMS/ETCS components, as the latter have to be developed ex novo. In particular, to better understand the impact of ERTMS/ETCS components' reliability on On-board system reliability, we performed a sensitivity analysis whose results are shown in Table 5 (with reference to a single On-board system). Row headings represent the scaling factors on variable parameters for the sensitivity analysis (e.g. Scale 0.1 for RTM means $MTBF^*_{RTM} = 0.1 * 10^6\ \text{h} = 10^5\ \text{h}$). The overall On-board system sensitivity to ERTMS/ETCS components' reliability is quite low when MTBF scales up or down of only one order of magnitude, as the EVC constitutes the main reliability bottleneck; when the reliability of ERTMS/ETCS components is scaled of two or more orders of magnitude, instead, the impact on $MTBF_{ONB}$ is more significant. By simply observing model structure, with the hypothesized reference architecture and in a Level 2 implementation, it does not appear to be any reason to assign a higher reliability to certain On-board components, as their influence only depends on their reference value and not on structural aspects (probably, the specification choice of differentiating them is related to the possibility for the On-board to "fall-back" into the lower Level 1, rarely implemented). Therefore, despite of component RAM specification, our system level analysis for an ERTMS/ETCS Level 2 implementation suggests a balanced choice of MTBF for ERTMS/ETCS components; e.g. all components' $MTBF = 10^6$ h, implying $MTBF_{ONB} = 6.3825 * 10^4$ h. Finally, Table 6 shows the impact of $MTTR_{ONB}$ and of the number of trains on overall On-board reliability and availability (only a selection of results is reported). Our analysis shows that the On-board MTBF requirements of Table 1 are not respected by our reference architecture, even with a low number of trains; however, such requirements are hardly fulfilled even by completely redundant

Table 6. On-board unavailability with respect to MTTR and number of trains.

| $MTTR_{ONB}$ | $N_{ONB}$ | $MTBF_{SF-ONB}$ | $MTBF_{IF-ONB}$ | $U_{SF-ONB}$ | $U_{IF-ONB}$ |
|---|---|---|---|---|---|
| 30′ | 2 | $3.1913*10^4$ | $9.5738*10^4$ | $1.5667*10^{-6}$ | $5.2223*10^{-6}$ |
| | 4 | $1.5956*10^4$ | $3.7232*10^4$ | $3.1335*10^{-5}$ | $1.3429*10^{-6}$ |
| | 6 | $1.0638*10^4$ | $2.3403*10^4$ | $4.6999*10^{-5}$ | $2.1364*10^{-5}$ |
| | 12 | $5.3188*10^3$ | $1.1121*10^4$ | $9.3997*10^{-5}$ | $4.4958*10^{-5}$ |
| | 24 | $2.6594*10^3$ | $5.4344*10^3$ | $1.8798*10^{-4}$ | $9.1998*10^{-5}$ |
| 1 h | 6 | $1.0638*10^4$ | $2.3403*10^4$ | $9.3994*10^{-5}$ | $4.2728*10^{-5}$ |
| 2 h | 6 | $1.0638*10^4$ | $2.3403*10^4$ | $1.8797*10^{-4}$ | $8.5452*10^{-5}$ |

On-boards using very reliable components. Therefore, they seem over dimensioned considering real EVC implementations (which constitute the limiting factor to reliability). Fortunately, from a system level point of view, it is sufficient to reason in terms of unavailability, whose results for the On-board are also reported in Table 6 and seem compatible with system level requirements which will be used in the global analysis of Section 6.

## 6 THE TRACKSIDE SUBSYSTEM

### 6.1 *Trackside model structure*

Most of the considerations already done about the architectural model of the EVC can be applied to the Radio Block Center, with the following two differences: (1) instead of On-board ERTMS/ETCS components, the RBC only features two communication interfaces (GSM-R and WAN); (2) the RBC is a repairable system, which can be maintained on-line by a dedicated technician. Therefore, while model structure remains substantially the same, the computation will be performed with respect to components' availability instead of MTBF. The Fault Tree formalism still suits such kind of analysis, in the infinite repair resources assumption: when a failure occurs to a component, the repair action starts immediately and finishes after a Mean Time To Repair which is independent from concurrent failures and does not account for possible system restart times (we assume them negligible; for more articulated maintenance policy modeling, refer to (Flammini 2005)). The RBC Fault Tree model is depicted in Figure 4. Just like the Lineside, the only failure mode for a RBC leads to an immediate system Immobilizing Failure, as the number of trains meant to be managed by each RBC is at least 2. Therefore, with respect to IFs, the Trackside can be modelled by a simple OR gate connecting all RBCs installed on the track.

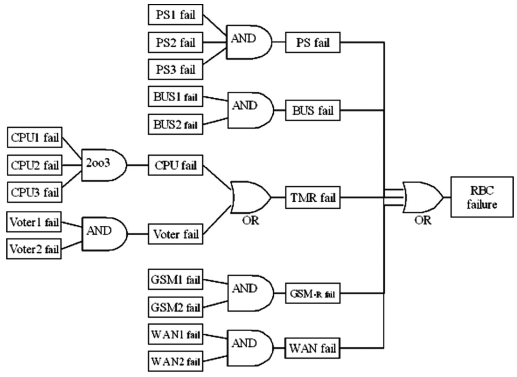



Figure 4. The Fault Tree model of the Radio Block Center.

Table 7. Trackside model parameters.

| Param. | Description | Value |
|---|---|---|
| $N_{RBC}$ | Total number of Radio Block Centres | 1-5 |
| $MTBF_{CPU}$ | MTBF of the Processor-Memory Card | $1.35*10^5$ h |
| $MTBF_{BUS}$ | MTBF of system Bus | $2.25*10^5$ h |
| $MTBF_{VOT}$ | MTBF of each FPGA based Voter | $3.33*10^8$ h |
| $MTBF_{PS}$ | MTBF of Power Supply | $5.50*10^4$ h |
| $MTBF_{GSM}$ | MTBF of GSM-R communication interface | $1.75*10^5$ h |
| $MTBF_{WAN}$ | MTBF of WAN communication interface | $4.00*10^5$ h |
| $MTTR_{RBC}$ | Mean time to replace a RBC component | 5′, 15′, 30′ |

### 6.2 *Trackside model parameters*

Refer to Section 2 for explanation about the COTS components used in the computing subsystem (the chosen MTBF are the same). For GSM-R and WAN interfaces, COTS components are used, too. The MTTR is assumed to be the same for all components, each of which is easy accessible and hot-replaceable. The MTTR variation set consists in typical values for supervised systems: 5, 10 and 30 minutes (the latter can correspond to a system with less easily accessible components or more hardly diagnosable faults).

Table 8. RBC unavailability with respect to repair times.

| $MTTR_{RBC}$ | $U_{RBC}$ |
|---|---|
| 1′ | $3.0182*10^{-7}$ |
| 5′ | $1.5145*10^{-6}$ |
| 5′ | $4.5454*10^{-6}$ |
| 30′ | $9.0909*10^{-6}$ |

Table 9. Trackside unavailability w.r.t. the number of RBCs.

| $N_{RBC}$ | $U_{IF-TRK}$ |
|---|---|
| 2 | $9.0909*10^{-6}$ |
| 3 | $1.3636*10^{-5}$ |
| 4 | $1.8182*10^{-5}$ |
| 5 | $2.2727*10^{-5}$ |

### 6.3 *Trackside model evaluation*

For the RBC, no MTBF requirement is given, so we can directly reason in terms of availability. Table 8 reports the evaluated unavailability of the Radio Block Center with respect to different repair times. Availability is related to reliability and maintainability according to the well know formula: A = MTBF/(MTBF + MTTR). Therefore, the result of strong dependence between $U_{RBC}$ and $MTTR_{RBC}$, shown in Table 8, is expectable and underlines the importance of adopting efficient repair strategies and hot-spare components: this allows satisfying the requirement on system availability ($U_{RBC} < 10^{-6}$) without using highly reliable and expensive ad-hoc components. However, for the system level analysis we won't consider the poorly realistic result corresponding to the lowest $MTTR_{RBC} = 1'$, as we will show that this is not necessary to satisfy the system level availability requirement. Finally, Table 9 shows the results about Trackside unavailability, assuming a realistic $MTTR_{RBC} = 15'$. According to the results obtained, the number of RBC should be kept as low as possible; however, other factors (e.g. performance requirements) constrain such a choice. As evaluated for the On-board (see Section 4), it could be shown that the requirement MTBF-$I_{TRK} > 3.5*10^8$ h is largely over-dimensioned: we will simply neglect it and proceed to our system level analysis.

## 7 THE GLOBAL MODEL OF HARDWARE FAILURES

### 7.1 *Global model structure*

For the global failure model, we decided to exploit the Bayesian Networks formalism as it allows to:

– model several failure modes (i.e. IF and SF) in a single model, by means of multi-state stochastic variables;

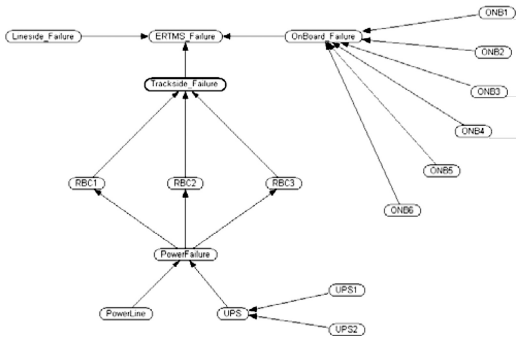



Figure 5. The global Bayesian Network model featuring a common mode failure.

– introduce and evaluate the system level impact of common mode failures, e.g. power failures;
– automatically locate system level criticalities, by a posteriori probabilities.

While these features can be separately provided by other formalisms, BN allow treating them in an integrated framework, and they do not suffer from the state space explosion problem.

The basic structure of the BN model (shown in Figure 5) is simply a translation of an omologous FT model, extended with the aforementioned specific features of BN. The ERTMS Failure event is modelled by a three state variable which represents the most significant ERTMS system level failures (IF, SF, MF or no failure), as described in Section 2. For instance, the Conditional Probability Table (CPT) for the "noisy" OR gate connected to ERTMS_Failure (a sort of Top Event for a Fault Tree) is shown in Table 10. As we can see from the CPT table, gate implementation is obtained by conditioning system failure probability to subsystems' failure probability, as described in more details in (Bobbio 2001b) (note that the On-board failure node is a three state event, as the On-board features two failure modes: Immobilising and Service).

The choice of modeling a common mode of failure is justified by the fact that in a real operating environment, all the RBC are located in the same building, in order to ensure easy maintenance, sharing the same power line. For the common source of failure to cause a system level failure, also the Uninterruptible Power Supplies (UPS) must fail, and such an event is modelled by a simple bayesian AND gate.

### 7.2 *Global model parameters*

The parameters of the final BN global model are no more varying in their full variability range, as assumed for previously described subsystems Fault Tree analyses, whose results have already been discussed above. Instead, they are chosen using the already available

Table 10. Conditional Probability Table of the noisy OR gate connected to the "Top Event".

| Lineside | Trackside | On-board | Immobilizing failure | Service failure | Minor or no failure |
|---|---|---|---|---|---|
| OK | OK | OK | No | No | Yes |
| OK | OK | KO_Immob | Yes | No | No |
| OK | OK | KO_Serv | No | Yes | No |
| Any other combination | | | Yes | No | No |

Table 11. Global model parameters.

| Param. | Description | Values |
|---|---|---|
| $U_{RBC}$ | RBC Unavailability | $1.5145*10^{-6}$,$4.5454*10^{-6}$, $9.0909*10^{-6}$ |
| $U_{EVC}$ | EVC Unavailability | $7.8339*10^{-6}$, $1.5668*10^{-5}$, $3.1335*10^{-5}$ |
| $U_{LNS}$ | Lineside Unavailability | $3.1959*10^{-7}$ |
| $U_{PWR}$ | Power Unavailability | $1.54*10^{-5}$ |
| $U_{UPS}$ | UPS Unavailability | $1.25*10^{-6}$ |

Table 12. A selection of system level results.

| Common cause | $U_{RBC}$ | $U_{EVC}$ | $U_{SF}$ | $U_{IF}$ |
|---|---|---|---|---|
| NO | $1.5145*10^{-6}$ | $7.8339*10^{-6}$ | **$4.4972*10^{-5}$** | **$5.7519*10^{-6}$** |
| (YES, | | $1.5668*10^{-5}$ | **$8.6449*10^{-5}$** | **$8.3687*10^{-6}$** |
| with redundant UPS) | | $3.1335*10^{-5}$ | $1.5956*10^{-4}$ | $1.8329*10^{-5}$ |
| | $4.5454*10^{-6}$ | $1.5668*10^{-5}$ | **$8.5664*10^{-5}$** | $1.7372*10^{-5}$ |
| | $9.0909*10^{-6}$ | $7.8339*10^{-6}$ | **$4.3956*10^{-5}$** | $2.8213*10^{-5}$ |
| | | $3.1335*10^{-5}$ | $1.5596*10^{-4}$ | $4.0507*10^{-5}$ |
| YES, with no UPS | $1.5145*10^{-6}$ | $7.8339*10^{-6}$ | **$4.4915*10^{-5}$** | **$6.9947*10^{-6}$** |
| | | $3.1335*10^{-5}$ | $1.5936*10^{-4}$ | $1.9556*10^{-5}$ |
| | $9.0909*10^{-6}$ | $3.1335*10^{-5}$ | $1.5576*10^{-4}$ | $4.1706*10^{-5}$ |

results and according to realistic assumptions about the number of trains (i.e. EVCs), RBCs and BGs, taken from real world system implementations and usage characteristics. In practical implementations, in fact, no more than 3 trains follow each other for each track direction, no more than 3 RBCs are used for each high-speed railway line, and Lineside results are related to high reliable balises used in groups of 2 (thus the Lineside subsystem is not even exploded in its basic components). Parameter values, meaning and variability range is reported in Table 11. UPS unavailability refers to high reliable and easily maintainable industrial models (e.g. $MTBF_{UPS} = 2 * 10^5$ h and $MTTR_{UPS} = 15'$); power line unavailability is assumed to be quite low with respect to normal users' perceptions for the usual presence of diesel generators which activate quickly in case of black-outs (e.g. $MTBF_{PWR} = 3$ months and $MTTR_{PWR} = 2'$).

### 7.3 *Global model evaluation*

First of all, a study can be performed on the model under analysis by exploiting the Most Probable Explanation of Bayesian Networks. If an Immobilising Failure occurs, the a posteriori failure probabilities are almost the 80% for the Trackside (about 26% for each RBC) and 16% for the On-board (nearly 6% for each system), therefore the former seems the main responsible for IFs (the Lineside contribution, once more, proves to be negligible). On the opposite side, when a Service Failure occurs, the responsibility is 100% allocated to the On-board, as expectable. The "sensitivity to findings" calculation provides and automated sensitivity analysis, in which the On-board branch gives the far higher contribution, suggesting the opportunity to act on On-board in order to improve system availability.

The results of global model evaluation are reported in Table 12. We can observe how the common mode failure contribution is negligible when its probability is kept low ($<10^{-9}$) by adding redundant UPS, while it is as more relevant as other components' unavailability decreases, partly annihilating the efforts made to design more available subsystems. The fundamental result is that the shaded cells of Table 12 highlight design choices fulfilling the system level

requirements:

- $U_{IF-HW} < 1.46 * 10^{-5}$ (from Table 1, $A_{IF-HW} > 0.9999854$ and obviously $U_{IF-HW} = 1 - A_{IF-HW}$), or
- $U_{SF-HW} < 1.30 * 10^{-4}$ (from Table 1, $A_{SF-HW} > 0.99987$ and obviously $U_{SF-HW} = 1 - A_{SF-HW}$).

The results in bold can be selected as valid design choices, as they fulfil both requirements on Immobilising and Service Failures. We recall that some of these results correspond to subsystems' MTBF which we showed in previous sections not to be compliant to ERTMS/ETCS RAM specification for constituents, and this underlines the value of a system level analysis (fulfilling the requirements for constituents would have been either unfeasible or too much expensive). Finally, the results also demonstrate how the use of properly redundant COTS components suits the engineering of high-available critical systems.

## 8 CONCLUSIONS AND FUTURE WORKS

In this paper we have shown a combined usage of Fault Trees and Bayesian Networks in order to evaluate system reliability aspects of the new European railway standard. In particular, for subsystems many results have been obtained by only relying on the Fault Trees, exploiting their flexibility and efficiency of analysis, while the global model analysis has been performed by means of Bayesian Networks, exploiting the enhanced modeling power of such formalism.

The analyses on ERTMS/ETCS presented in this paper allowed us to obtain several useful results. First of all, we showed the advantages of a system level analysis with respect to a one based on constituents: the former allows using less reliable (e.g. COTS) components and fulfil system reliability requirements at a lower cost. Secondly, we highlighted some incoherence in reliability requirements stated by the specification (some values are over-dimensioned with respect to other ones). Last but not least, we were able to find out optimal design choices in order to fulfil reliability requirements since early design stages, only basing on the specification and on the proposed reference architecture. The compositional approach and the combination of Fault Tree and Bayesian Network formalisms revealed their advantages in terms of power and flexibility in performing the presented study.

We are currently evaluating the possible advantages of expressing the whole model by means of BN, also considering advanced dynamic (Montani 2005) and decision extensions. Decision Networks (also known as Influence Diagrams) allow to evaluate system-level cost-benefit design trade-offs: we will try to augment the power of analysis of BN using their decisional extensions, namely decision and utility nodes (see e.g. (Watthayu 2004)). Decision extensions can be exploited to perform automated cost-benefit analyses on input reliability parameters of the model (e.g. MTBF, redundance level, etc.). System cost raises with components' number (linearly) and reliability (exponentially), while benefits include system performance and availability. More complex dependencies arise if we consider the impact of maintenance costs, which are obviously lower for a system with a limited number of more reliable components (at equal availability).

## REFERENCES


Ajmone Marsan, M.; Balbo, G.; Conte, G.; Donatelli, S. & Franceschinis G. 1995. *Modeling with Generalized Stochastic Petri Nets*: J. Wiley.

Bobbio, A.; Bologna, S.; Ciancamerla, E.; Franceschinis, G.; Gaeta, R.; Minichino, M. & Portinale, L. 2001a. Comparison of Methodologies for the Safety and Dependability Assessment of an Industrial Programmable Logic Controller. *Proceedings of ESREL 2001*, Torino.

Bobbio, A.; Portinale, L.; Minichino, M. & Ciancamerla, E. 2001b. Improving the Analysis of Dependable Systems by Mapping Fault Trees into Bayesian Networks. *Reliability Engineering and System Safety Journal* – 71/3: pp. 249–260.

CENELEC 1999. *EN 50126 Railways Applications – The specification and demonstration of Reliability, Maintainability and Safety (RAMS)*.

Dugan, J.B.; Bavoso, S.J. & Boyd, M.A. 1992. Dynamic Fault-Tree Models for Fault Tolerant Computer Systems. *IEEE Transactions on Reliability*, vol. 41, 1992: pp. 363–377

Flammini, F.; Iacono, M.; Marrone, S. & Mazzocca, N. 2005. Using Repairable Fault Trees for the evaluation of design choices for critical repairable systems. *Proceedings of the 9th IEEE International Symposium on High Assurance Systems Engineering* (HASE2005), Heidelberg, Germany, October 12–14: pp. 163–172.

Montani, S.; Portinale, L.; & Bobbio A. 2005. Dynamic Bayesian Networks for Modeling Advanced Fault Tree Features in Dependability Analysis. *Proc. of European Safety and Reliability Conference* (ESREL 2005), Tri City, Poland: pp. 1415–1422.

NASA Office of Safety and Mission Assurance 2002. *Fault Tree Handbook with Aerospace Applications*, ver. 1.1

Netica web site 2006: http://www.norsys.com/netica.html

Portinale, L.; Bobbio, A. & Montani, S. 2004. From AI to Dependability: Using Bayesian Networks for Reliability Modeling and Analysis. *Proceedings of the Fourth International Conference on Mathematical Methods in Reliability* (MMR2004).

Sahner, R.A.; Trivedi, K.S. & Puliafito, A. 1996. *Performance and Reliability Analysis of Computer Systems: An Example-based Approach Using the SHARPE Software Package*: Kluwer Academic Publishers.

UNISIG 1999. *ERTMS/ETCS – RAMS Requirements Specification*, Ref. 96s1266l.

UNISIG 2002. *ERTMS/ETCS – Class1 SRS Issue 2.2.2, Subset-026*.

Watthayu, W. et al. 2004. A Bayesian network based framework for multi-criteria decision making. *Proceedings of the 17th International Conference on Multiple Criteria Decision Analysis*.

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